

Van-Thinh To

Paris, France | van-thinh.to@etu.u-paris.fr | vanthinh.to.github.io | [Google Scholar](#)
linkedin.com/in/thinhthovan | github.com/VanThinhTo

EDUCATION

Erasmus Mundus Scholar - Double Degree Master's Program

University Paris Cité (France)

Master of Bioinformatics

GPA: 18.05/20.00 (90%)

2025 - expected 2027

University of Milan (Italy)

Master of Chemistry

GPA: 18.05/20.00 (90%)

2025 - expected 2027

University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam)

Bachelor of Pharmacy (5-year program)

GPA: 3.31/4.00 (83%)

2019 - 2024

Rank top 4%

Thesis: A Knowledge-Guided Graph Self-Supervised Learning for Molecular Property Predictions

Grade: 10/10

RESEARCH EXPERIENCE

University Paris Cité (France)

Supervisor: Prof. Olivier Taboureau and Assistant Prof. Viet-Khoa Tran-Nguyen

Master's researcher student

1. Large-scale PubChem/ChEMBL bioactivity curation and virtual screening benchmarking Sep 2025 - present

- Automating large-scale retrieval of bioactivity data for all protein targets from PubChem and ChEMBL.
- Identifying dark chemical matter compounds for use as assumed inactives.
- Conducting virtual screening benchmarking studies across selected protein targets.

2. Target-specific machine learning scoring functions for cystathionine β -synthase inhibitors Jan 2026

- Curated data set from PubChem and ChEMBL.

University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam)

Supervisor: Assoc. Prof. Tuyen Ngoc Truong and PhD. Researcher Tieu-Long Phan

Undergraduate researcher

1. Generative AI for dual targets of EGFR and VEGFR May 2024 - August 2025

- Developed a knowledge-guided graph-based scoring function to evaluate generated molecules.

2. A knowledge-guided graph for molecular representation Dec 2023 - Jul 2024

- Introduced pre-text tasks for self-supervised learning guided by orbital information.
- Proposed a novel knowledge-guided graph (KGG) utilizing a hierarchical molecular graph, defragmentation, and two knowledge vectors that incorporate orbital information for molecular representation.
- A 2.4% improvement in ROC-AUC for classification tasks, along with a 39%, and 14% reduction in RMSE, and MAE respectively for regression tasks, compared to a state-of-the-art model on the MoleculeNet dataset.

3. A novel application of machine learning in virtual screening of HIV integrase inhibitors Jun 2022 - Jan 2024

- Identified the applicability domain of the machine learning models using Principal Component Analysis for dimension reduction, coupled with convex hull techniques.

4. Application of molecular similarity, neural network and GNINA molecular docking in virtual screening of PD-L1 inhibitors Jun 2023 - Dec 2023

- Proposed an artificial neural network model, and a molecular similarity assessment; utilized GNINA 1.0 molecular docking.
- Developed an automated pipeline for the molecular similarity model.
- Screened 15,235 DrugBank repurposing compounds and identified 22 hit compounds.

5. Machine learning and Deep neural networks with consensus molecular docking for enhanced potency prediction of ALK inhibitors Dec 2022 - Jun 2023

- Trained an ensemble model including a graph neural network, an artificial neural network and a XGBoost model for the prediction of potential Anaplastic Lymphoma Kinase inhibitors.

SELECTED PUBLICATIONS

Full publication list

1. **To, V.-T.**, Van Nguyen, P.-C., Truong, G.-B., Phan, T.-M., Phan, T.-L., Fagerberg, R., Stadler, P. F. & Truong, T. N. KGG: Knowledge-Guided Graph Self-Supervised Learning to Enhance Molecular Property Predictions. *Journal of Chemical Information and Modeling*. doi:10.1021/acs.jcim.5c01068 (2025)
2. Van Nguyen, P.-C., **To, V.-T.**, Tran, N.-V. N., Phan, T.-L., Truong, T. N., Gärtner, T., Merkle, D. & Stadler, P. F. SynCat: molecule-level attention graph neural network for precise reaction classification. *Digital Discovery*. doi:10.1039/D5DD00367A (2026)
3. **To, V.-T.**, Phan, T.-L., Doan, B.-V. N., Nguyen, P.-C. V., Le, Q.-H. N., Nguyen, H.-H., Trinh, T.-C. & Truong, T. N. Innovative virtual screening of PD-L1 inhibitors: the synergy of molecular similarity, neural networks and GNINA docking. *Future Medicinal Chemistry*, 1–12. doi:10.1080/17568919.2024.2389773 (2024)
4. Phan, T.-M., Phan, T.-L., Van-Nguyen, P.-C., Le, H. S. L., **To, V.-T.**, Truong, T. N., Merkle, D. & Stadler, P. F. ProQSAR: A Modular and Reproducible Framework for Small-Data QSAR Modeling with Fit-and-Use Models. *Journal of Cheminformatics*. doi:10.1186/s13321-026-01175-9 (2026)
5. Phan, T.-L., Trinh, T.-C., **To, V.-T.**, Pham, T.-A., Van Nguyen, P.-C., Phan, T.-M. & Truong, T. N. Novel machine learning approach toward classification model of HIV-1 integrase inhibitors. *RSC advances* **14**, 14506–14513. doi:10.1039/D4RA02231A (2024)
6. Truong, G.-B., Pham, T.-A., **To, V.-T.**, Le, H.-S. L., Nguyen, V. P.-C., Trinh, T.-C., Phan, T.-L. & Truong, T. N. Discovery of VEGFR-2 Inhibitors employing Junction Tree Variational Encoder with Local Latent Space Bayesian Optimization and Gradient Ascent Exploration. *ACS Omega*. doi:10.1021/acsomega.4c07689 (2024)
7. Trinh, T.-C., Phan, T.-L., **To, V.-T.**, Pham, T.-A., Truong, G.-B., Le, L. H. S., Tran, X.-T. D. & Truong, T. N. Synergy of advanced machine learning and deep neural networks with consensus molecular docking for virtual screening of anaplastic lymphoma kinase inhibitors. *Journal of Computer-Aided Molecular Design* **39**, 79. doi:10.1007/s10822-025-00657-6 (2025)

TEACHING EXPERIENCE

University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam)

Aug 2024 - Aug 2025

Contracted teaching assistant and research mentoring - Department of Organic Chemistry, Faculty of Pharmacy

- Wrote research proposals for four university-funded projects, each awarded 1200 USD.
- Prepared training materials on computer-aided drug design for undergraduate students, such as RDKit, fingerprints, molecular descriptors, ANN, GNN.
- Mentored one undergraduate student on thesis project focused on self-supervised learning.
- Taught practical organic chemistry, with a focus on laboratory equipment proficiency, ChemDraw usage, lab techniques and synthesis skills.

SKILLS

- **Cheminformatics:** RDKit; Machine Learning (Scikit-learn); Deep learning (TensorFlow, PyTorch); Molecular Docking (GNINA); Generative diffusion model (IRDiff); Property-matched decoy generation (DeepCoy); Databases (PDB, PubChem, ChEMBL); Visualization (PyMOL)
- **Programming skills:** Programming language (Python, R, C), Linux terminal, Pytest, Git, Package management, deployment
- **IELTS:** Overall 7.0 (Academic Version)

Nov 2024

TRAINING

Vienna Summer School on Drug Design (Austria)

Sep 2023

- Gained foundational training in drug discovery using docking, pharmacophore and molecular simulation tools (LigandScout, KNIME, OpenEye)
- Learned concepts of techniques in drug discovery, such as generative models.

TALKS

1. **To, V.-T.**, Van-Nguyen, P.-C., Truong, G.-B., Phan, T.-M., Phan, T.-L., Fagerberg, R., Stadler, P. & Truong, T. KGG: Knowledge-Guided Graph Self-Supervised Learning to Enhance Molecular Property Predictions in 41st Scientific and Technical Conference of Pharmacy (2025)(Oral - Presenter)
2. **To, V.-T.**, Phan, T.-L. & Ngoc Doan, B.-V. Application of Molecular Similarity and Artificial Neural Networks for PD-L1 inhibitors Virtual Screening in ECMC2023: 9th International Electronic Conference on Medicinal Chemistry (2023) (Poster - Presenter)

SELECTED AWARDS

- Pharmacy Scholarship for Elite Students** 2019 - 2024
Awarded a \$5000 stipend over 5 years, covering full tuition (acceptance rate: 10%)
- Second prize in the nationwide Vietnamese Chemistry Olympiad** 2018, 2019
The biggest Chemistry competition for high school students in Vietnam with 400+ participants, ranking 42 overall
- Odon Vallet Scholarship** 2019, 2024
Awarded to outstanding high school, university, and graduate students in Vietnam.

REFERENCES

Associate. Prof. Tuyen Ngoc Truong University of Medicine and Pharmacy at Ho Chi Minh city (Vietnam) truongtuyen@ump.edu.vn	PhD. Researcher Tieu-Long Phan University Leipzig (Germany) tieu@bioinf.uni-leipzig.de
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